

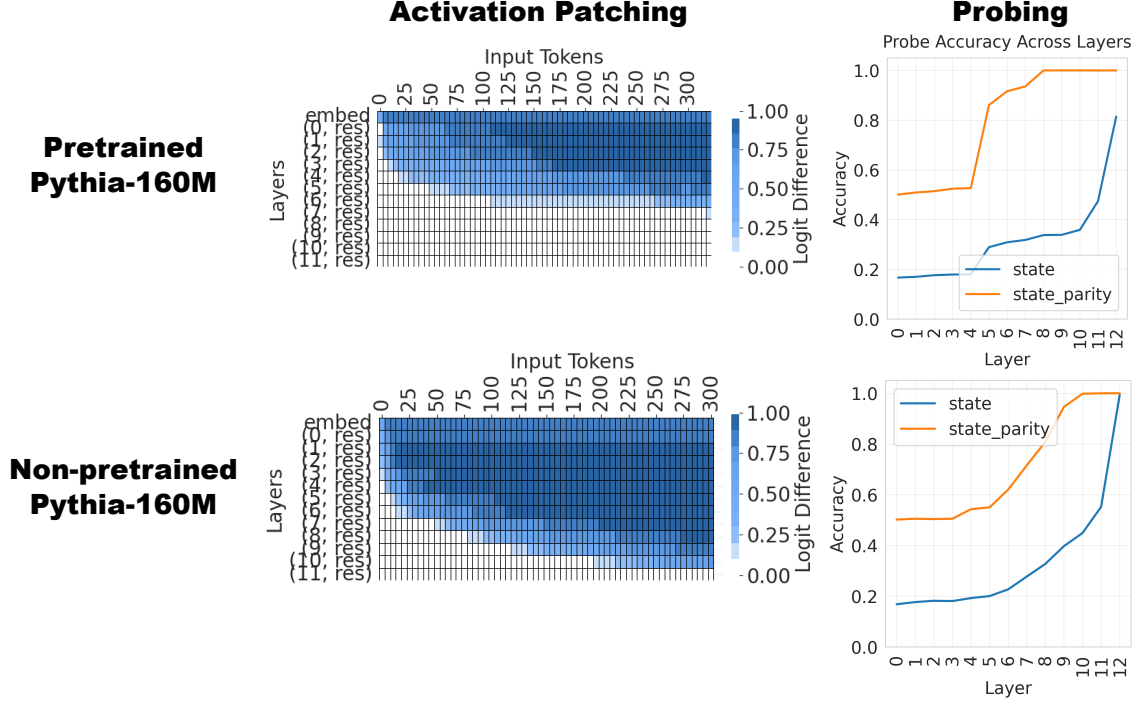


Figure 14. Patching and probing result for Pythia models trained on *natural language* permutation composition task. We plot the signature of a pre-trained Pythia model (top) and a non-pre-trained Pythia model (bottom). In both cases, the signatures are consistent with the state being learned associatively (both the patching signature and state probe have an exponential curve). The signature of the pre-trained model is consistent with a PAA signature, with the parity probe converging in early layers, and the activation patching signature containing a light-blue middle section.

Define the sets of attention weights on odd and even tokens of x as:

$$\mathcal{A}_{\ell,H,\text{odd}}(x) = \{\alpha_{i,\ell}^H(x) : a_i \text{ is odd}\},$$

$$\mathcal{A}_{\ell,H,\text{even}}(x) = \{\alpha_{i,\ell}^H(x) : a_i \text{ is even}\}.$$

We find heads that respond to parity can be local: for example, attention head 5 in layer 1 responds to odd-parity permutations in the midpoint of the sequence, while attention head 4 responds to ones late in the sequence.

Thus, we record the **parity head score** $\sigma_{\ell,H,\text{parity}}$, which measures the *proportion* of the sequence for which more attention is placed on odd-parity actions compared to even-parity actions:

$$\sigma_{\ell,H,\text{parity}} = \frac{1}{L-5} \cdot \sum_{t=5}^L \sigma_{\ell,H,\text{parity},t}, \quad \text{where}$$

$$\sigma_{\ell,H,\text{parity},t} = \mathbf{1} \left(\mathbb{E}[\mathcal{A}_{\ell,H,\text{odd}}([a_1 \dots a_i])] - 0.95 \cdot \text{CI}_{\ell,H,\text{odd}}([a_1 \dots a_i]) > \mathbb{E}[\mathcal{A}_{\ell,H,\text{even}}([a_1 \dots a_i])] \right).$$

Here, $[a_1 \dots a_i]$ denotes the first i elements of the sequence x , and $\text{CI}_{\ell,H,\text{odd}}$ refers to the confidence interval around the average attention weights on odd-parity tokens. We use sequence lengths of up to $L = 80$ for S_3 and $L = 50$ for S_5 .

We find no evidence of parity heads in any layer of AA models. However, we find at least two attention heads with σ significantly exceeding 50% in the first few layers of PAA models, highlighted in Table 2.

F.2. AA Attention Patterns

What sorts of attention patterns appear in AA models? Because attention is dense, we visualize only the top-K attention traces from and to each position at each layer. Specifically, we plot the attentions of an LM on an input as a graph with:

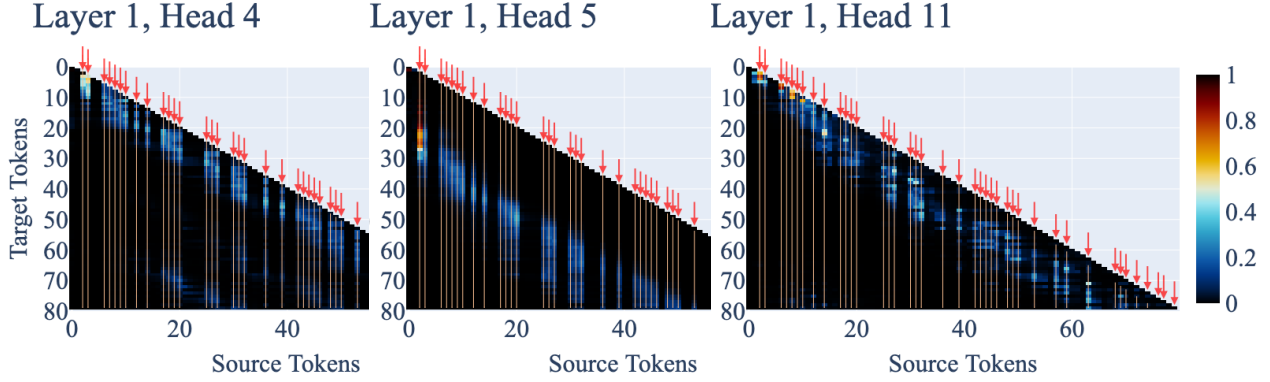


Figure 15. Examples of parity heads in a PAA model. We plot heatmaps showing attention weights on each source token at each target token location (we show up to only 60 source tokens for heads 4 and 5 for the sake of space). We draw arrows / yellow lines at source tokens corresponding to *odd-parity actions*. Note that parity heads attend almost exclusively to odd-parity tokens.

Algorithm	Layer	Head	Parity Head Score
S_3 (PAA)	1	1	90.1% $\pm$ 3.0%
	1	4	86.4% $\pm$ 8.2%
	0	5	67.1% $\pm$ 5.1%
S_3 (AA)	0	4	3.3% $\pm$ 4.5%
	2	7	3.0% $\pm$ 6.1%
	11	0	2.0% $\pm$ 4.0%
S_5 (PAA)	3	3	83.6% $\pm$ 9.6%
	3	2	80.6% $\pm$ 6.2%
	2	6	50.3% $\pm$ 22.4%
S_5 (AA)	0	7	5.9% $\pm$ 8.2%
	0	3	3.8% $\pm$ 5.4%
	0	0	3.2% $\pm$ 4.3%

Table 2. Top-3 parity head scores across all attention heads in each type of model. We report average parity head scores (%) over 100 examples, as well as their standard deviations. Informally, this metric captures the proportion of the sequence over which more attention is placed on odd-parity actions than even-parity actions.

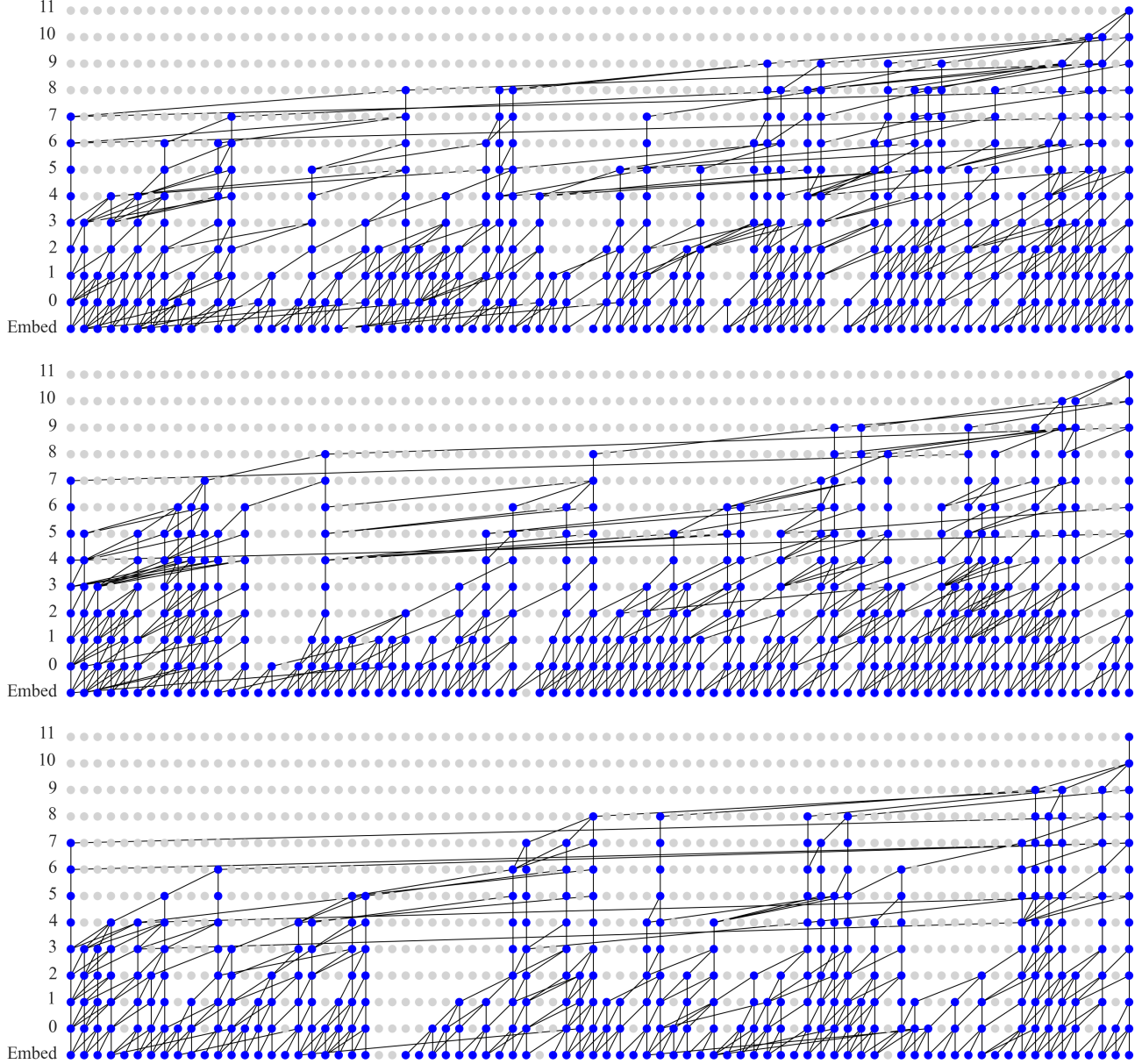


Figure 16. Attention patterns in AA models form a tree-like pattern, with tokens in successive layers attending to larger windows of downstream tokens. This is in line with how we expect the associative algorithm to function. We plot only the most salient attention weights by pruning edges for which the attention weight is < 0.95 , the target token is not in the top-3 attended-to tokens from the source token, or the source token is not in the top-10 attended-from tokens for the target token. We expect that the attention patterns visualized here do not form a single clean tree, but the superimposition of multiple trees.

1. Nodes (t, l) for each token position t and layer l ,
2. Edges between two nodes $(t_1, l - 1)$ and (t_2, l) if position t_2 at layer l attends to position t_1 at layer $l - 1$. We define “attends to” as follows:

Let $\alpha_{t_1 \rightarrow t_2}^{(l)}$ denote the maximum attention weight (across all attention heads at layer l) from position t_1 at layer $l - 1$ to position t_2 at layer l .

We say (t_2, l) **attends to** $(t_1, l - 1)$ if all of the following conditions are met: